



Unit 4 · Outcome 1 · Climate Change

Can we slow it down? Mitigation cuts climate change off at the source.

KK11

VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.



01What “net” actually means

01 What “net” actually means The dot point doesn’t say “reduce emissions”. It says reduce net emissions — and that one word is the whole concept. Examiners love it because most students skim past it. Mitigation means taking action to reduce the net amount of greenhouse gas in the atmosphere , so that climate change is slowed or limited. It attacks the cause of the enhanced greenhouse effect (covered in KK04): too much heat-trapping gas. “Net” emissions are a balance — a subtraction: Σ The net-emissions equation $\text{NET emissions} = \text{sources} - \text{sinks}$ Sources add

02Mitigation vs adaptation — the distinction examiners test most

02 Mitigation vs adaptation — the distinction examiners test most This is the single most common trap in the whole Climate Change outcome. Get it crisp and you bank easy marks; blur it and you lose them. Mitigation KK11 Adaptation KK12 Targets the... cause — the greenhouse gas itself effects — the impacts we can’t avoid Goal Reduce net emissions to slow / limit warming Build resilience and cope with the warming already locked in Scale of benefit Global — a tonne avoided anywhere helps everywhere (CO_2 is well-mixed) Local — protects a particular place or co

	Mitigation KK11	Adaptation KK12
Targets the...	cause — the greenhouse gas itself	effects — the impacts we can’t avoid
Goal	Reduce net emissions to slow / limit warming	Build resilience and cope with the warming already locked in
Scale of benefit	Global — a tonne avoided anywhere helps everywhere (CO_2 is well-mixed)	Local — protects a particular place or community
Timeframe of benefit	Delayed — decades, due to gas lifetime & ocean inertia	Can be immediate for the protected area
Examples	Solar & wind power; EVs; reforestation; methane-cutting cattle feed; carbon capture	Sea walls; drought-tolerant crops; cyclone-rated buildings; relocating infrastructure; early-warning systems

03The mitigation toolkit

03 The mitigation toolkit There is no single fix. Real decarbonisation stacks many options across every sector — “silver buckshot, not a silver bullet.” Group them by whether they shrink a source or grow a sink . Shrink the sources — emit less source Decarbonise electricity Replace coal & gas generation with solar, wind, hydro (and nuclear elsewhere). The single biggest lever, since electricity is a huge slice of emissions. source Electrify & be efficient Swap gas heaters, petrol cars and industrial burners for efficient electric equivalents (heat pu

04Net-zero ledger lab

04 Net-zero ledger lab Drive Cape Wirreanda’s region to net zero. Pull the mitigation levers and watch the bathtub: shrink the orange inflow, grow the green outflow, and hold the water level steady. Net = sources – sinks. Net-emissions bathtub 3D view unavailable — use the readouts and toggles below; they work without graphics. Sources (Mt/yr) 50.0 Sinks (Mt/yr) 8.0 Net (Mt/yr) 42.0 Net positive — the atmosphere is still filling. Keep going. 95% renewable electricity –18 Electrify & efficiency –8 Electric transport –7 Cut livestock methane –5 Reforestati

05 Stacking the wedges

05 Stacking the wedges Another way to picture it: a growing gap between “business as usual” emissions and a safe pathway. No single action closes it — each option is a wedge , and you stack wedges until the gap is gone. Stabilisation wedges 3D view unavailable — toggle the wedges below to see the gap close. Gap remaining: 100% — switch wedges on to close it. Renewable electricity Energy efficiency Electrified transport Forests & land carbon Methane reduction Capture & storage The lesson of the wedges Each wedge is achievable on its own, but only the st

06 Australia & Victoria — real targets, real projects

06 Australia & Victoria — real targets, real projects VCAA rewards specific, local, current examples. Memorise a couple of these and deploy them as evidence. Level Target How they’re mitigating Australia Net zero by 2050 (legislated, Climate Change Act 2022); 43% below 2005 by 2030; 62–70% by 2035 Net Zero Plan’s five priorities: clean electricity, electrification & efficiency, clean fuels, new technologies, and net carbon removals; the Safeguard Mechanism caps big emitters Victoria Net zero by 2045 ; 75–80% below 2005 by 2035; 95% renewable electricity

Level	Target	How they’re mitigating
Australia	Net zero by 2050 (legislated, Climate Change Act 2022); 43% below 2005 by 2030; 62–70% by 2035	Net Zero Plan’s five priorities: clean electricity, electrification & efficiency, clean fuels, new technologies, and net carbon removals; the Safeguard Mechanism caps big emitters
Victoria	Net zero by 2045 ; 75–80% below 2005 by 2035; 95% renewable electricity by 2035	VRET renewables target, 6.3 GW storage by 2035, 9 GW offshore wind by 2040, the State Electricity Commission, ZEV uptake

07 Evaluating mitigation options

07 Evaluating mitigation options When a question says evaluate , discuss or justify , listing options isn’t enough. Weigh each one against criteria — strengths and limitations. Criterion Ask yourself... Effectiveness How much net abatement does it deliver? Is the gas it tackles high-warming-potential (KK06)? Cost Cheap and proven (solar, wind, efficiency) or expensive and emerging (direct air capture)? Feasibility / maturity Ready to deploy now, or still developing? Does it need new infrastructure or storage? Co-benefits Cleaner air, jobs, biodiversity, en

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Co-benefits	Cleaner air, jobs, biodiversity, energy security, healthier soils?
Limitations / trade-offs	Land competition, intermittency, lifecycle (embodied) emissions, permanence risk, social impact on communities?
Timeframe	Benefits are delayed for everyone — so urgency and the precautionary principle matter.

08 Sort it: mitigation or adaptation?

◇ ACTIVE RECALL

08 Sort it: mitigation or adaptation? Drag each action into the right bin. The test: does it remove or avoid greenhouse gas (mitigation), or just help us cope with impacts (adaptation)? Mitigation — cuts net emissions
Shrinks a source or grows a sink Adaptation — manages effects Builds resilience to unavoidable change s

09 Exam technique

09 Exam technique Decode the command word, then structure with P·E·L. Compare a weak script to a strong one and copy the moves that earn marks. Command-word decoder Identify Describe Explain Distinguish Compare Evaluate Justify Discuss P·E·L — structure every extended answer P Point State the option or claim directly. Name the mitigation strategy and the sector it sits in. E Evidence How it cuts net emissions (source or sink), plus a specific Australian/Victorian example or figure. L Link Tie back to the question — “slowing climate change”, the limitatio

10 Practice questions

★ PRACTICE & SELF-MARK

10 Practice questions VCAA-style, escalating from 2 to 6 marks. Attempt each before revealing. Self-mark honestly — the dock tracks your total. Q1 Define mitigation in the context of climate change, and explain what the word “net” means in “reducing net greenhouse emissions.” 2 marks ► ★ Model answer Mitigation is action taken to reduce net greenhouse gas emissions in order to slow or limit climate change by addressing its cause. “Net” means the balance after subtracting removals by sinks from emissions by sources (net = sources – sinks), so net emission

★ Good vs bad worked answers

Study the contrast — the same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer structure — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

er script 1 / 4 Mitigation and adaptation are both ways to deal with climate change . Mitigation is when you stop climate change and adaptation is when you get used to it . For example you can build sea walls and use solar power . Why 1/4: no real distinction drawn (cause vs effect never stated); “stop climate change” overclaims; examples aren’t matched to the right term — the sea wall is actually adaptation but isn’t labelled. Only the solar reference earns credit.

✓ STRONG / MODEL ANSWER

er script 4 / 4 Mitigation reduces net greenhouse emissions to address the cause of climate change, whereas adaptation builds resilience to the effects we can no longer avoid. For example, switching to wind power is mitigation because it cuts emissions at the source; building a sea wall against rising seas is adaptation because it manages an impact without removing any greenhouse gas. Why 4/4: explicit cause-vs-effect contrast (the heart of “distinguish”), one correctly-labelled example each, and each example tied back to why it belongs to that category.

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Worked good/weak answers illustrate exam technique — always check against current VCAA marking advice.